Paleomagnetism of the Precambrian Martin Formation, Saskatchewan

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Oriented samples of the Martin Formation were collected from 23 sites, both sediments and igneous rocks being represented. Detailed laboratory treatment yields an overall mean direction corresponding to a paleomagnetic pole at 9° S, 72° W (dp = 6.3°, dm = 11.5°). Strong evidence for the primary nature of the magnetization involved is provided by a positive fold test. The paleomagnetic pole obtained provides further evidence for a major eastward loop in the polar wandering path for North America during late Aphebian – early Helikian times. The exact age of Martin Formation deposition is the subject of debate, but extreme limits given by various authors are 1930 to 1490 m.y. However, the paleomagnetic results themselves somewhat restrict this range, and are of use in setting up a regional chronology, since they enable stratigraphic correlations between isolated sedimentary basins of the Canadian Shield to be made.

A vingt-trois endroits de la formation de Martin, on a pris des échantillons orientés aussi bien dans les roches sédimentaires qu'ignées. Une étude détaillée en laboratoire donne une direction moyenne du pôle paléomagnétique de 9° S, 72° ouest (dp = 6.3°, dm = 11.5°).

Il y a une forte présomption que la magnétisation soit primaire, un essai de plissement ayant donné un résultat positif. La détermination de ce pôle paléomagnétique fournit une autre preuve d'une boucle vers l'est au cours de la migration du pôle pour l'Amérique du Nord à la fin de l'Aphébien ou au début de l'Hélikien. L'âge précis des dépôts de la formation de Martin fait l'objet de la discussion bien que les limites extrêmes établies par différents auteurs soient de 1930 à 1490 millions d'années. Les résultats paléomagnétiques obtenus restreignent ces limites et peuvent être utilisés dans l'établissement d'une chronologie régionale puisqu'ils permettent de corrélations stratigraphiques entre des bassins sédimentaires isolés dans le Bouclier Canadien. [Traduit par le journal]

Introduction

Knowledge of the early history of the geomagnetic field is still rather rudimentary. This fact is in part a necessary consequence of the fragmentary nature of the Precambrian geological record. Nevertheless, there do exist in the world areas of essentially unmetamorphosed strata, which can be expected to yield important data concerning early geomagnetic behavior and large-scale tectonism. In particular, there exist in the northwestern Canadian Shield a number of Proterozoic basins within which are preserved thick sequences of redbeds and lavas (Fraser *et al.* 1970). This paper reports paleomagnetic investigations from one of the basins.

The sampling area is located in the Beaverlodge area of northwestern Saskatchewan (Fig. 1) and the materials sampled are red sediments, lavas, and sills of the Martin Formation. A detailed description of the area has been given by Tremblay (1968). Briefly, the Martin Formation consists of at least 13 000 ft (3960 m) of conglomerate, arkose, siltstone, basaltic to andesitic flows, and gabbro sills, preserved in a small, fault-bounded basin regarded by Fraser *et al.* (1970) as a taphrogeosyncline. The clastic sediments are unmetamorphosed, but Tremblay (1968) pointed out that the volcanic rocks are generally highly altered (probably deuterically).

The major feature of the area is the Martin Lake syncline, the limbs of which attain dips exceeding 80°. Stratigraphically the Martin Formation overlies the highly metamorphosed Tazin Group with a pronounced angular unconformity. Numerous radiometric dates have been obtained from Tazin Group rocks, but a Rb/Sr whole-rock isochron on pegmatites cutting the Tazin but not the Martin is of most significance here. This isochron yielded an age of 1975 ± 20 m.y. (Sassano *et al.* 1972), which is a firm uppermost limit for Martin sedimentation. Fraser *et al.* (1970) adopted a maximum age of 1835 ± 50 (K–Ar on biotite) obtained from a basic dike that cuts the Tazin but "does not seem to cut rocks of the unmetamorphosed Martin Formation" (p.

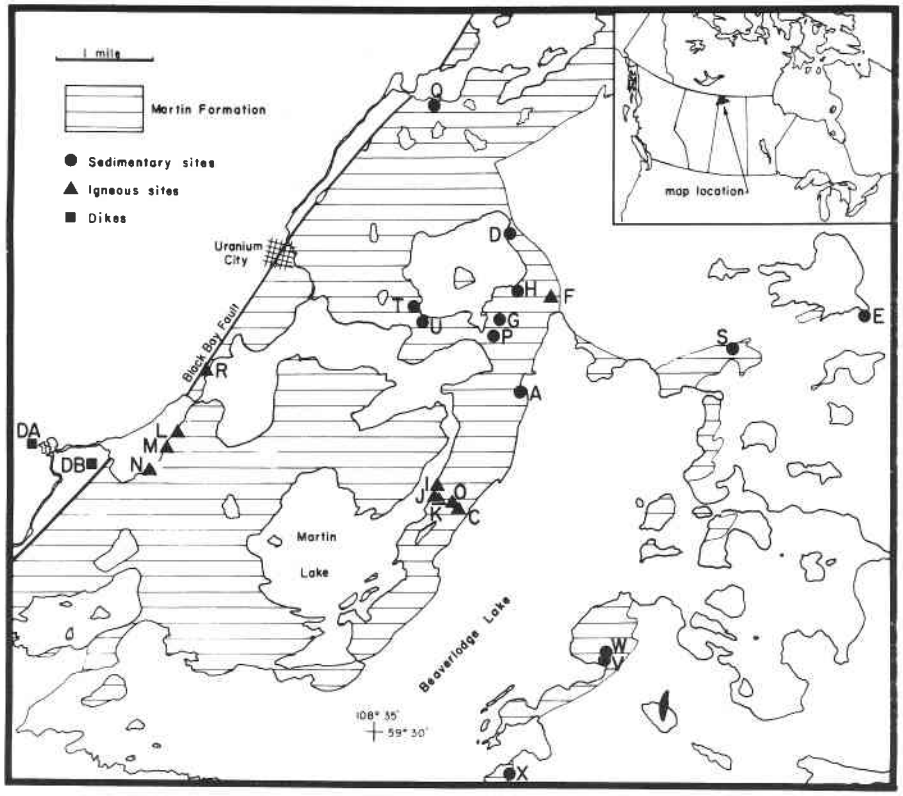


FIG. 1. Map of sampled area (modified after Tremblay (1968)). The Martin Formation site names used in the text are preceded by an "M", but this is omitted here for clarity. Site ME was collected in a mine some 2500 ft (762 m) below the surface.

217). No younger rocks directly overlie the Martin, but a gabbro dike cutting the lower horizons has been dated at 1490 ± 100 m.y. (K–Ar whole-rock). One of the basaltic flows of the Martin Formation has yielded a K–Ar whole-rock age of 1635 ± 180 m.y. In their summary, Fraser *et al.* (1970) regarded the Martin Formation as having formed between 1830 and 1650 m.y. ago; however, Koeppel (1968) using U–Pb data on two episodes of mineralization preferred limits of 1930 ± 40 m.y. to 1780 ± 20 m.y.

Methods

Samples were collected from a total of 23 sites, 13 of which were situated in sediments and 10 in igneous rocks. At each of the sedimentary sites 3 or 4 hand samples were collected, yielding a total of 40 blocks. The igneous samples were collected by both field-drilling (36 cores) and hand-sampling (13 blocks). Most samples were oriented using a sun-compass. Due to overcast conditions some sedimentary samples were oriented magnetically. Comparison of magnetic and solar bearings, where possible, indicates that orientation errors do not exceed 5°.

All measurements of remanence were made on a commercial spinner magnetometer and routine partial demagnetization studies have been carried out on all specimens used in the final analysis. Alternating field (a.f.) demagnetization was performed with an apparatus closely similar to that described by McElhinny (1966). Thermal cleaning experiments employed a small furnace located in a room-sized magnetic shield similar to that described by Patton and Fitch (1962); the magnetic field over the volume occupied by the specimens is not more than 5γ.

The procedure adopted involves step-wise partial demagnetization (by both methods) of one pilot specimen per sample and analysis of the results in terms of within-site scatter. The

treatment yielding minimum scatter has then been chosen for a second specimen from each sample, subject to the availability of material. This procedure closely resembles that proposed by Irving *et al.* (1961); however, we have also required that the individual pilot specimens show evidence of 'end-points' in the sense described by McElhinny and Gough (1963). The entire study involved approximately 1800 measurements of remanence made on about 250 specimens cut from 89 separately oriented samples. In the statistical analysis of the results unit weight is given to samples.

Results

Due to the widely different response to partial demagnetization of the igneous and sedimentary samples, these two groups are conveniently discussed separately.

(a) Sedimentary Sites

Thermal and magnetic cleaning has been carried out on representative specimens from every site, and in general the thermal results are more acceptable. This is in accord with most published results from redbeds containing hematite, and arises from the very high coercivities typical of this mineral, which often lie beyond the capacity of alternating field equipment.

With one exception (site MQ), the NRM's of all sites are non-randomly grouped according to Watson's (1956) criterion, although in a few cases the circle of confidence (α_{95}) includes the present geomagnetic field at the sampling locality, indicating that the rocks have probably been remagnetized in the direction of the present geomagnetic field (see Table 1). At all non-random sites, both thermal and magnetic cleaning result in systematic movement away from the present field direction; thermal treatment continues this movement beyond the a.f. 'end-points', and thus appears more effective in removing secondary components. In addition, most sites are more tightly grouped after thermal than after magnetic cleaning (Table 1). For these reasons the thermally cleaned site means are considered to be the better estimates of the ancient geomagnetic field directions and are consequently used in obtaining the overall paleomagnetic pole position. However, for site ME the magnetically cleaned result is used, as in this one instance thermal treatment leads to poor results ($k = 6.2$), which are not significantly different from the present field, whereas a.f. treatment yields well-grouped end-points ($k = 53.2$) away from the present geomagnetic field. The reason for this behavior is not understood.

Specimens from all 13 sedimentary sites respond to standard laboratory techniques for the removal of unwanted secondary magnetizations. Eleven of the site means form a group, containing both polarities, around an axis NW-and-up/SE-and-down (Fig. 3). Discussion of this overall mean is postponed until after the description of the results from igneous sites. The two remaining sites (MP and MQ) are thought to represent genuine intermediate directions acquired whilst geomagnetic polarity reversals were taking place. They are more conveniently discussed in section (*c*).

(b) Igneous Sites

Only five of the ten sites sampled yielded acceptable data. The five unacceptable sites are discussed first. Three of these have been completely remagnetized by the present field (MC, MF, ML; Table 2); pilot specimens subjected to a.f. and thermal treatment failed to give any indication of an underlying primary magnetization. Specimens from the other two unacceptable sites (MJ and MN) have variable behavior and do not yield good site means. Some (from MJ) are strongly magnetized and do not respond at all to either thermal or a.f. treatment and have probably been subject to lightning strikes; others (from both sites) possess large soft components of magnetization and never reach acceptable end-points. Both sites are poorly grouped at all stages of treatment. These five sites (MC, MF, MJ, ML, MN) cannot be regarded as possessing acceptable records of the ancient geomagnetic field and are therefore eliminated from further discussion.

Laboratory treatment of specimens from the remaining five sites successfully removed secondary components and led to site means in agreement with the results obtained from the sediments. Of these five, one site (MK) possesses both polarities and is discussed in more detail below. The remaining four sites (MI,

TABLE 1. Paleomagnetic results for sedimentary sites of Martin Formation (after correction for geological dip)

	NRM							Cleaned					
Site	*N*	*S*	*D*	*I*	*k*	*R*	*T*/*H*	*N*	*S*	*D*	*I*	*k*	*R*
MA	3	3	031.7	+83.4	53.1	2.9623	630 °C	3	3	**160.9**	**+45.3**	108.2	2.9815
	3	3	074.8	+79.6	42.9	2.9533	1000 Oe	3	3	161.2	+61.3	22.3	2.9104
MD	3	3	329.4	+28.8	10.1	2.8017	600 °C	3	6	**324.7**	**−14.0**	30.2	2.9337
	3	3	321.0	+23.8	18.9	2.8941	1000 Oe	3	3	323.5	+04.5	16.0	2.8747
ME	3	5	(161.0	+52.6)	26.3	2.9238	630 °C	3	6	(133.5	+34.1)	6.2	2.6760
	3	6	(122.0	+41.6)	9.2	2.7838	800 Oe	3	6	**145.3**	**+40.3**	53.2	2.9624
MG	4	8	(322.1	+12.2)	23.2	3.8709	550 °C	4	8	**314.9**	**−16.0**	99.5	3.9698
	4	4	316.7	+13.4	59.2	3.9493	1400 Oe	4	4	313.2	+05.6	47.0	3.9361
MH	3	3	305.9	+25.9	34.0	2.9411	500 °C	3	6	**305.6**	**+07.3**	24.4	2.9179
	3	3	308.1	+20.6	49.7	2.9597	1000 Oe	3	3	306.7	+15.8	24.0	2.9166
MP	4	4	344.4	+60.0	5.7	3.4691	660 °C	4	4	214.6	+71.0	12.8	3.7659
	4	4	314.5	+44.2	7.5	3.6014	1400 Oe	4	4	264.5	+45.0	15.8	3.8100
MQ	3	3		*	2.2	2.1055	thermal	3	3		*		
	3	3		*	2.5	2.2018	a.f.	3	3		*		
MS	2[a]	2	(148.8	+41.5)	38.4	1.9740	600 °C	2[a]	4	**138.4**	**+28.4**	201.4	1.9950
	2[a]	4	(143.8	+40.4)	20.6	1.9513	800 Oe	2[a]	4	138.5	+33.2	37.6	1.9734
MT	4	4	108.4	+36.8	41.6	3.9279	500 °C	4	4[b]	**139.2**	**+36.7**	178.1	3.9832
	4	8	112.1	+40.6	74.6	3.9598	800 Oe	4	8	130.0	+36.6	126.3	3.9762
MU	4	4	041.4	+38.9	22.7	3.8678	600 °C	4	8	**341.9**	**−25.1**	36.1	3.9168
	4	4	(059.0	+33.8)	24.2	3.8762	1400 Oe	4	4	014.1	+06.3	15.9	3.8116
MV	3	3	329.9	−23.9	36.9	2.9458	500 °C	3	6	**333.4**	**−40.1**	123.9	2.9839
	3	6	334.1	−28.9	105.1	2.9810	1400 Oe	3	6	337.0	−36.3	235.3	2.9915
MW	3	3	330.0	−45.2	115.5	2.9827	600 °C	3	6	**330.8**	**−50.1**	84.0	2.9762
	3	6	332.1	−45.2	50.4	2.9603	1000 Oe	3	6	331.9	−48.7	42.5	2.9529
MX	3	3	316.7	−40.0	14.0	2.8572	550 °C	3	6[b]	**327.3**	**−51.4**	28.6	2.9301
	3	6	318.7	−36.7	16.6	2.8793	600 Oe	3	6	319.4	−41.1	15.9	2.8740

N = number of samples; *S* = number of specimens; *D*, *I* = declination, inclination of site mean direction; *k* = precision parameter (Fisher 1953); *R* = resultant of *N* unit vectors; *T*/*H* = temperature/peak field treatment.
Bold-face type indicates data used in calculating overall mean direction.
Parentheses indicate directions whose α_{95} (Fisher 1953) include the present geomagnetic field (before correction for geological dip).
*Directions random on Watson's (1956) criterion.
[a]One otherwise well-behaved sample rejected because of widely divergent direction; orientation error suspected.
[b]Due to lack of material one or more specimens thermally treated after partial a.f. demagnetization to 1000 Oe or more.

MM, MO, MR) require no special comments beyond the information given in Table 2.

(c) Sites with Intermediate Directions

Three sites possess magnetic vectors lying between the two polarity groups formed by the main body of the reliable data reported here. The details are summarized in Fig. 2. The behavior during demagnetization of individual specimens is similar to that observed at other sites and gives no reason to doubt the paleomagnetic reliability of the remanence. However, the directions indicated by the end-points often diverge widely from the group formed by the other sites. It seems likely that these anomalous sites have recorded the geomagnetic field at times when it was actually reversing. Consideration of the stratigraphic relationships between these sites (discussed in more detail below) suggests that three separate reversals are involved, and this is consistent with the different nature of the results from each of the three sites. At site MQ (sediment), pilot specimens exhibit very good end-points spread along a great circle passing close to the overall mean direction. Data from site MP (sediment) are a little more scattered than most sites (Table 1), but the site mean is definitely not random. It lies approximately half way between the two polarity groups. Finally, specimens from site MK (igneous) apparently possess components of *both* polarities. During thermal demagnetization all three pilot specimens move systematically towards the 'normal'[1] polarity group, but

[1]'Normal' and 'reversed' are used in the sense defined by Reid (1972), namely, that polarity which yields north poles lying on the polar wander curve traceable to the present north pole is termed normal.

TABLE 2. Paleomagnetic results from igneous sites of Martin Formation (after correction for geological dip). Results from two nearby dikes are also included

	NRM							Cleaned					
Site	*N*	*S*	*D*	*I*	*k*	*R*	*T*/*H*	*N*	*S*	*D*	*I*	*k*	*R*
MC	3	3	(315.2	+45.4)	11.3	2.8223	400 Oe thermal[a]	3	3	(323.4	+39.4)	58.9	2.9661
MF	3	3	*		2.1	2.0541	200 Oe thermal[a]	3	3	(266.9	+18.6)	41.1	2.9514
MI	8	8	312.2	+10.8	10.7	7.3474	630 °C	8	8	**315.2**	**−25.5**	32.4	7.7843
	8	8	303.2	−10.1	3.8	6.1716	1000 Oe	8	8	319.3	−12.7	18.6	7.6245
MJ	8	8	(325.8	+05.9)	2.2	4.7941	thermal				*		
	8	8	*		1.6	3.7281	a.f.				*		
MK	3	3	(336.1	+23.0)	16.7	2.8804	660 °C a.f.[b]	3	3		*	1.2	1.3080
ML	4	4	(076.1	+45.0)	7.2	3.5844	300 Oe thermal[a]	4	4	(082.5	+41.7)	186.8	3.9839
MM	3	3	(098.8	+23.6)	49.1	2.9593	550 °C	3[c]	3	105.7	+19.7	77.6	2.9742
	3	3	106.6	+07.7	59.1	2.9661	100 Oe	3[c]	3	**113.1**	**+21.6**	268.3	2.9925
MN	5	5	(095.6	+07.2)	5.0	4.1965	a.f.[d] thermal[d]						
MO	8	8	(327.9	+33.9)	18.3	7.6167	500 °C	8	8	308.8	+09.6	8.5	7.1789
	8	8	(325.3	+31.1)	9.6	7.2745	400 Oe	8	8	**303.1**	**−19.2**	25.7	7.7275
MR	2	4	(153.4	+08.6)	1.4	1.3052	600 °C	2	4[e]	**175.8**	**+18.4**	14.7	1.9320
	2	2	(151.2	+20.7)	2.4	1.5865	500 Oe	2	2	(164.5	+24.1)	13.4	1.9255
Dikes													
DA	3	3	171.4	+37.7	41.0	2.9513	400 °C	3	3	169.2	+32.9	147.5	2.9864
	3	3	161.9	+41.8	71.5	2.9720	150 Oe	3	3	162.3	+33.0	151.2	2.9868
DB	2	2	181.1	+65.8	167.9	1.9940	500 °C	2	2	181.2	+64.2	455.2	1.9978
	2	2	145.0	+73.1	22.6	1.9557	150 Oe	2	2	149.1	+67.4	21.4	1.9532

Symbols have same meaning as in Table 1, except for the following:
[a]All samples remagnetized in present field; one pilot thermal demagnetization gave same result as a.f. treatment, therefore thermal treatment not pursued further.
[b]No a.f. 'end-points' observed.
[c]Two samples (out of 5) rejected because of lack of 'end-points'.
[d]Unstably magnetized (see text).
[e]Two specimens previously magnetically cleaned to 1800 Oe.

above 550 °C two of them shift rapidly to stable end-points near the 'reversed' direction.

The obviously complex nature of the remanence at these three sites indicates that it would be unwise to include their site means in the overall mean direction for the entire formation.

(d) Basic Dikes

Two basic dikes thought to be associated with the Martin Formation volcanism were sampled for preliminary work. These two dikes lie outside the main taphrogeosynclinal basin of the Martin Formation proper, but detailed mapping by Tremblay (1968) indicated that similar dikes cut all formations in the area up to the Martin Formation volcanic rocks. It is possible, therefore, that these dikes are feeders for the Martin igneous rocks and as such can be expected to yield paleomagnetic data similar to the Martin rocks. Results from the two dikes sampled (sites DA, DB) appear to favor this interpretation (Fig. 3, Table 2), but the data available at present are not sufficient to allow a firm conclusion. Further work is planned to elucidate the time relationships between these dikes and the Martin Formation.

Discussion

The fifteen acceptable site mean directions appear in Tables 1 and 2 in bold-face type and are plotted in Fig. 3; they possess the following parameters: $D = 322.5$, $I = -29.4$, $\alpha_{95} = 10.4°$, $k = 14.5$ (for the purposes of calculating this mean direction the five directions lying in the SE quadrant have been reversed). Before corrections for geological tilt these site means possess considerably greater scatter, with $k = 3.3$. Application of McElhinny's (1964) test indicates that this increase in scatter is

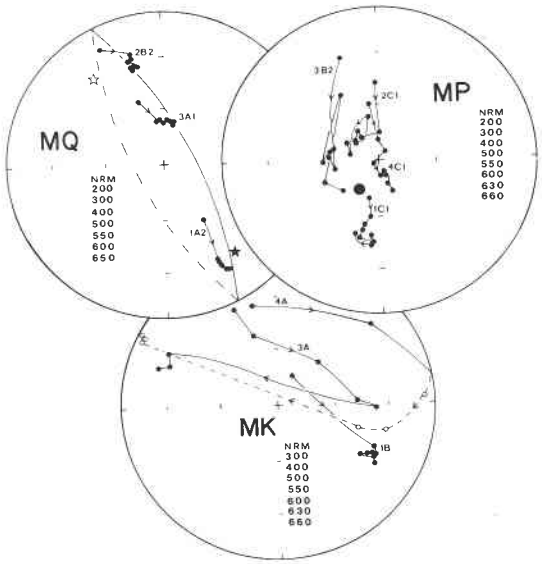


FIG. 2. Thermal demagnetization of specimens with intermediate directions. The treatments used (°C) are indicated. The star represents the overall mean direction for the Martin Formation. The great circle was fitted by inspection. The large dot represents the site mean direction for site MP. Equal-area stereographic projections; solid (open) symbols represent north-seeking directions on lower (upper) hemisphere.

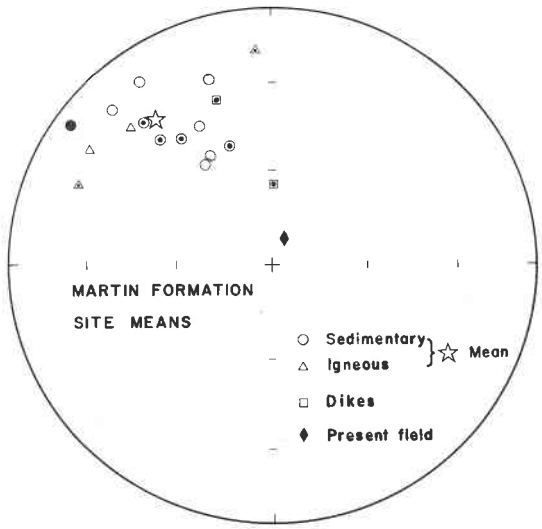


FIG. 3. Mean site directions of magnetization for the Martin Formation. Equal-area stereographic projection. Solid (open) symbols represent north-seeking directions on the lower (upper) hemisphere; open symbols with central dots represent south-seeking directions on the upper hemisphere (see Tables 2 and 3).

significant at the 99% level, and this, in turn, implies that the magnetization pre-dates the folding. Tremblay (1968) concluded that the bulk of the folding took place more or less immediately after deposition (see his Fig. 17). The data therefore yield a positive fold test (Graham 1949), which demonstrates that the paleomagnetic pole obtained genuinely reflects the geomagnetic field in Martin Formation times.

The scatter of the site means is often taken as a measure of the paleosecular variation. For this purpose the precision of the site means freed from within-site scatter is appropriate. A standard two-tier analysis (Watson and Irving 1957) yields values of 19.8 and 40.5 for between-site (k_B) and within-site (k_W) precision respectively. The angular standard deviation, $\Delta = 81/k_B^{1/2}$, is thus 18.2°, and has 95% confidence limits such that $14.6° \leqslant \Delta \leqslant 24.1°$ (Cox 1969). Brock (1971) has analyzed paleomagnetic data for all ages in terms of Cox's (1970) geomagnetic model, and obtains a best fitting value of 17.1° for the paleolatitude in question (16°). Thus the intrinsic scatter in the Martin Formation site means is typical of that expected from secular variation; it is also statistically indistinguishable from the value obtained by Creer (1962) from an analysis of the present geomagnetic field (epoch 1945).

On the assumption of a geocentric dipole field, the overall mean direction corresponds to a paleomagnetic pole at 9° S, 72° W (dp = 6.3°, dm = 11.5°), which is plotted in Fig. 4, along with other relevant data from North America. This result indicates that northern Saskatchewan was situated well within the tropics (paleolatitude = 16°) during deposition of the Martin Formation, which concurs with the findings of Briden and Irving (1964) that 88% of redbed occurrences for which paleomagnetic control exists formed at paleolatitudes less than 30°. Earlier geological work led to the suggestion that the Martin sediments formed under hot conditions (Alcock 1936; Christie 1953; Blake 1956), but Tremblay (1968) favored a cold climate because of the unaltered nature of the feldspars. He also argued that the presence of large faceted fragments of Tazin rocks at various horizons in the Martin suggested that ice may have been

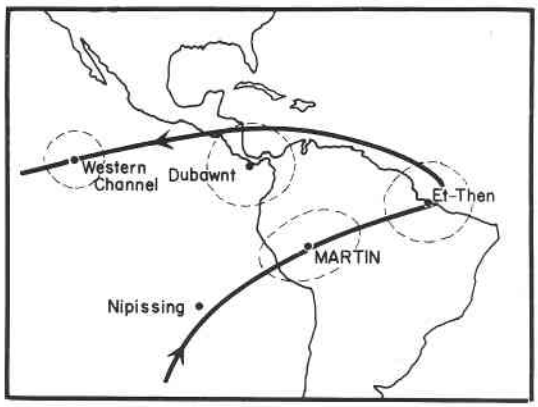


FIG. 4. Part of the polar wander curve for North America (after Irving and Park (1972)), with relevant paleomagnetic poles and their 95% confidence ovals. The Nipissing Diabase pole is that given by Symons (1970). Since only one sill is represented this result must be regarded as preliminary, and the oval of confidence is omitted to emphasize this fact. More recent data (Symons 1971) from Nipissing sheets in a separate locality are highly scattered, but yield a similar pole position. Details of other results can be found in the following references: Et–Then Group (Irving *et al.* 1972*a*), Dubawnt Group (Park *et al.* 1973), Western Channel Diabase (Irving *et al.* 1972*b*).

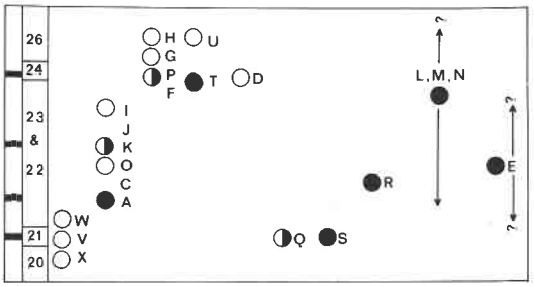


FIG. 5. Approximate stratigraphic relationship between sites and implied geomagnetic polarity sequence. The numbers refer to map-units defined by Tremblay (1968). The figure is schematic and no thickness or time scales are implied. Vertical columns represent sites that are stratigraphically well ordered with respect to each other. No horizontal scale or ordering is intended. Open (closed) symbols represent reversed (normal) polarity, half-filled symbols are intermediate directions (see text). Sites are identified by single letters (*i.e.* the prefix 'M' is omitted). The stratigraphic position of site E (collected underground) and sites L, M, and N (sills) is uncertain.

involved in their transportation. This seems unlikely on the basis of the paleomagnetic evidence which strongly favors a low latitude environment.

As noted above, the results indicate the occurrence of one or more geomagnetic polarity reversals. It is therefore pertinent to enquire what information about the temporal behavior of the ancient geomagnetic field is contained in the data. Precise stratigraphic control over all the sites is not available; however, the relationship within certain groups of sites is definitely known, and Tremblay's (1968) excellent detailed map enables the relative positions of these groups to be roughly established. The available information is summarized in Fig. 5. It appears that the intermediate directions recorded at sites MQ and MP are related to the polarity intervals represented by sites MS and MT respectively. But the intermediate directions at site MK are not associated with any observed normal interval. The normal direction at site MR may be associated with either MA or MK; stratigraphic uncertainty does not permit exact correlation. The information displayed in Fig. 5 suggests that at least four normal periods occurred, but on the whole the field seems to have been mainly reversed.

Earlier versions of the polar wander path relative to North America (Spall 1971; Brock and Piper 1972) showed a simple northerly track in the eastern Pacific during late Aphebian time. More recently, however, Irving *et al.* (1972*a*) have modified the curve to include a major eastward loop over northern South America. This new loop is of similar magnitude to, but older than, the "Logan Loop" described by Robertson and Fahrig (1971). Recently, Irving and Park (1972) have argued that the sharp changes in direction of apparent polar motion associated with these loops are of considerable tectonic significance, representing times when major global tectonic reorganization occurred. The paleomagnetic pole deduced from the Martin Formation falls on "track 5" of Irving and Park's polar wander curve and lies between the poles obtained from the Nipissing diabase (2162 ± 27 m.y. (Fairbairn *et al.* 1969; Symons 1970, 1971)) and the Et–Then Group (bracketed by 1250 and 1845 m.y., Irving *et al.* (1972*a*)). It appears that the Martin Formation pre-dates the Et–Then Group, which, in turn, predates the Dubawnt Group (1720–1735 m.y., Park *et al.* (1973)). As discussed above, currently avail-

able radiometric ages for the Martin Formation have extreme limits of 1930 to 1490 m.y., but in their recent summary Fraser *et al.* (1970) adopted limits of 1830 to 1650 m.y. If the sequence of poles implied by Irving and Park's curve is valid, then the age of Martin sedimentation is further constrained to exceed about 1730 m.y. However, it is stressed that the pattern of polar movement during this early period of earth history is still poorly defined and future work may well modify preliminary conclusions drawn at this stage.

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